MirrorMe: LuminaView Interactive Smart Mirror Using Raspberry Pi

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ABSTRACT

MirrorGrid is a locally-hosted smart mirror operating system that delivers real-time, gesture-driven navigation across eight information panels without requiring cloud connectivity or physical touch input. The system employs a decoupled client-server architecture: a Python FastAPI backend provides a REST data API and a WebSocket telemetry stream, while a single-page browser frontend integrates Google MediaPipe Hands for in-browser gesture classification, Three.js r128 for GPU-accelerated WebGL particle rendering, and GSAP for UI animation. Phase 5.1 resolves a camera hardware conflict between the Python OpenCV vision pipeline and the browser-side MediaPipe stack, introduces context-sensitive habit tracker gesture overrides, and aligns the system clock with IST (UTC+5:30) for Tamil Nadu deployment. Formal acceptance testing conducted under ISO/IEC 25010 confirms sub-2 ms warm-path REST latency (μ = 1.36 ms, σ = 1.07 ms), sub-50 ms gesture-response latency (overall mean = 11.08 ms), and 100% state machine correctness over 16 sequential transitions. Frame-rate benchmarking demonstrates 25–35 fps on the Raspberry Pi 5 target platform under WebGL GPU acceleration, exceeding the 24 fps human visual persistence threshold. All three acceptance tests return PASS, qualifying the system for research publication and physical deployment.

Keywords: smart mirror, gesture recognition, MediaPipe Hands, FastAPI, Three.js, WebSocket, Raspberry Pi, WebGL, IoT, real-time systems, ISO/IEC 25010

1.Introduction

Smart ambient displays have attracted growing research interest as a natural-language and gesture-driven human-computer interaction paradigm. Unlike handheld devices, mirror-form-factor displays are embedded in everyday environments — bedrooms, gyms, healthcare settings — and must operate without keyboards, mice, or touchscreens. MirrorGrid addresses this niche by implementing a fully local, gesture-controlled information dashboard that runs on commodity single-board computer hardware.

The system presents eight thematically distinct information panels: a home clock/cheat-sheet, a daily weather and agenda brief, a live commute map, a priority email digest, real-time hardware telemetry, an AI-curated world news feed, an interactive daily habit tracker, and a finance snapshot. Navigation between panels is achieved exclusively through natural hand gestures classified by Google MediaPipe Hands running in-browser via

WebAssembly, eliminating network round-trips for gesture data and minimising latency.

Phase 5.1, documented here, resolves a critical camera hardware conflict introduced in Phase 5.0, extends gesture vocabulary with context-sensitive habit-tracker actions, and refines the UI for high-contrast legibility on mirror substrates. Formal acceptance testing under ISO/IEC 25010 Quality Requirements accompanies this report, along with a two-tier frame-rate benchmark characterising rendering performance on both desktop and Raspberry Pi 5 target hardware.

1.1Motivation and Scope

The core engineering challenge for ambient mirror displays is latency: any perceptible delay between a user's gesture and the system's response degrades the illusion of a reactive mirror. Nielsen's HCI guidelines classify responses under 100 ms as "instantaneous"; sub-50 ms responses are not consciously registered as delayed at all [2]. MirrorGrid's architecture is specifically engineered to meet this threshold on low-power hardware, making it suitable for production deployment on a Raspberry Pi 5 without a dedicated GPU beyond the integrated VideoCore VII.

This report covers Phase 5.1 changes and their associated acceptance evidence. Sections 2 and 3 describe system architecture and implementation. Section 4 presents the formal acceptance test results. Section 5 presents the frame-rate benchmark. Section 6 discusses limitations and future work.

2.System Architecture and Design

2.1Overview

MirrorGrid is built on a decoupled, asynchronous Client-Server architecture. The two major subsystems are a Python FastAPI backend and a single-file Vanilla JS / Three.js frontend. The separation is deliberate: the browser is the authoritative source of truth for UI state and gesture decisions, while the server is responsible only for hardware telemetry and static data hydration. This prevents the common race condition in distributed mirror OS designs where stale server broadcasts overwrite locally-committed gesture decisions.

State Variable	Authoritative Source	Propagation	Update Rate
sleeping / page	Browser (local JS)	DOM mutation only	Instantaneous
cpu / ram	Server (psutil)	WebSocket broadcast	50 ms (20 Hz)
time / weather	Server (asyncio)	WebSocket broadcast	50 ms (20 Hz)
gesture / pointer	Browser (MediaPipe)	Local execution + POST	~30 fps (per frame)

Table 1. State authority model — Browser-authoritative design prevents sleep/page race conditions.

2.2Backend: FastAPI and Python

The server component (main.py) is implemented in Python 3.13 using FastAPI 0.111.0 served by Uvicorn 0.29.0. It exposes three primary endpoints:

- GET /api/data — Returns a static JSON payload containing mock weather forecast, task list, email digest, and traffic data, used to hydrate the UI at boot.
- POST /api/gesture/{gesture} — Receives gesture commands from the frontend and updates the server-side mirror state machine, providing a log-synchronised replica of browser state.

- WebSocket /ws/stream — Maintains a persistent, full-duplex connection. An asyncio background task polls live CPU and RAM metrics via psutil every 50 ms and broadcasts a JSON payload to all connected clients.

The backend state machine manages two orthogonal state dimensions: sleep/wake status and the current active page (0–7). Both dimensions are replicated from the browser to the server via REST POST calls; the server does not push page or sleep state to the browser, preventing desynchronisation.

2.3Frontend: Vanilla JS, Three.js, and MediaPipe

The frontend is a single self-contained index.html file. Its rendering stack layers native HTML/CSS over a hardware-accelerated Three.js WebGL canvas (r128) configured with 3,000 point-sprite particles forming a Saturn-ring-inspired ambient background. Particle rotation is driven by the requestAnimationFrame loop at whatever rate the GPU sustains, with a live FPS counter displayed on the System Stats panel.

MediaPipe Hands loads asynchronously from a CDN as a WebAssembly binary (~10 MB, cached after first load). Once initialised, it processes each webcam frame (640 × 480 at 30 fps) to produce 21 three-dimensional hand landmarks, which the detectGesture() function maps to one of six canonical gesture classes (Table 2).

Gesture	Debounce / Trigger	Action
CLOSED_FIST	800 ms hold	Sleep mode
OPEN_PALM + Wave	≥ 2 reversals / 800 ms	Wake system
THUMB_UP	1200 ms hold	Next page
POINTING (swipe)	Δx > 18% screen	Page left / right
PEACE / V-sign	1500 ms hold	Jump to News (Page 5)
OPEN_PALM (static)	Per-frame	Particle tracking; reset swipe

Table 2. Gesture vocabulary — finger presence encoded as [thumb, index, middle, ring, pinky].

2.4Technology Stack

Table 3 summarises all software components and their roles within the system.

Component	Version	Role
FastAPI	0.111.0	Backend REST API + WebSocket server
Uvicorn	0.29.0	ASGI server, production-grade async I/O
psutil	5.9.8	Host CPU and RAM telemetry polling
MediaPipe Hands	CDN latest	21-landmark ML hand detection (WASM)
Three.js	r128	WebGL particle scene, GPU render loop
GSAP	3.9.1	UI panel animations and particle tweens

Component	Version	Role
Leaflet.js	1.9.4	Interactive commute map (Google tile layer)
Python	3.13	Backend runtime

Table 3. MirrorGrid Phase 5.1 technology stack

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3.Phase 5.1 Implementation Updates

3.1Camera Hardware Conflict Resolution

A significant defect was identified in Phase 5.0 wherein the browser frontend was unable to access the system webcam. Root-cause analysis revealed that vision_tracker.py — launched by main.py at startup — was capturing the camera feed via OpenCV, creating an exclusive OS-level lock on the device node (/dev/video0 on Linux; the equivalent DirectShow binding on Windows 11). A webcam cannot be simultaneously accessed by two independent OS processes.

The resolution adopted in Phase 5.1 was to set VISION_AVAILABLE = False in main.py, unconditionally disabling the OpenCV tracker at startup. The vision_tracker.py module is retained in the codebase for optional future re-enablement (e.g., via an environment variable flag) but is excluded from the boot sequence. The frontend now holds exclusive camera access and runs MediaPipe gesture detection locally in WebAssembly with no change to its implementation.

3.2Context-Sensitive Habit Tracker Gestures

Phase 5.1 introduces context-aware gesture overrides when the active page is Page 6 (Daily Habits). Two new gesture actions are defined:

- THUMB_UP (held > 1200 ms on Page 6): Finds the topmost incomplete habit entry in the DOM, marks it as DONE with a green checkmark, and triggers a progress-bar recalculation with an animated fill transition.
- THUMB_DOWN (held > 1200 ms on Page 6): Finds the topmost incomplete habit entry and marks it as SKIPPED with a red icon, likewise recalculating the animated progress bar.

The override mechanism inserts a page-context guard at the entry point of applyGestureLocal() before the global gesture dispatch table is evaluated. This preserves all pre-existing gesture semantics on all other pages while enabling richer per-page interaction on Page 6.

Additionally, CLOSED_FIST sleep triggering was decoupled from a passive resting-hand detection failure mode. In Phase 5.0, any frame classified as CLOSED_FIST — including frames where the user's hand was simply relaxed at rest — could trigger an unintended sleep transition. Phase 5.1 mitigates this by requiring the CLOSED_FIST classification to persist continuously for a 800 ms debounce window before any state change is committed.

3.3UI Aesthetics and Legibility Improvements

Three targeted aesthetic improvements were applied:

- Background dimming: The opacity of the Three.js particle canvas (#three-canvas) was reduced to 25%, allowing the ambient background to remain visually present without competing with foreground information cards.
- Card contrast: Panel card backgrounds were deepened to rgba(10, 15, 22, 0.85) with a 30 px black box-shadow

(rgba(0,0,0,0.8)). Combined with the neon typography, this achieves a high contrast ratio suitable for reflective mirror substrates under ambient lighting.

- Typography: Agenda and to-do list timestamps were refonted from the geometric Orbitron typeface to Inter, improving legibility at small point sizes where Orbitron's wide stroke geometry reduces x-height clarity.

3.4Real-World Data Integration

Phase 5.1 replaces the generic OpenStreetMap tile layer with Google Maps Standard street view tiles for the commute dashboard (Page 2). The previous CSS colour-inversion filter applied to the map container was removed, allowing the map to render in accurate, full-colour cartography.

The JavaScript Intl.DateTimeFormat engine is now hardcoded to the Asia/Kolkata timezone (UTC+5:30), overriding any host OS or browser locale timezone. This eliminates the clock drift previously observed on Tamil Nadu deployments where certain browser builds defaulted to UTC or a different regional offset.

3.5WebSocket Connection Correction

The frontend WebSocket connection URI was corrected from a legacy endpoint to ws://localhost:8000/ws/stream. The backend asyncio telemetry loop was updated to broadcast CPU and RAM metrics unconditionally on every 50 ms tick (previously, the broadcast was conditional on client subscription state, which caused missed frames on reconnection). The frontend JSON parser was updated to handle the unified {type: "telemetry", cpu: …, ram: …, time: …, weather: …, traffic: …} payload structure.

4.Acceptance Testing and Results

Formal acceptance testing was conducted on 6 May 2026 against the ISO/IEC 25010:2011 Systems and Software Quality Model [1]. Tests were executed by an automated PowerShell harness (Antigravity AI) performing black-box HTTP REST and state-inspection testing against the local FastAPI server (http://localhost:8000) on a Windows 11 x64 host.

4.1Test 1 — REST API Response Latency

Objective

Measure end-to-end HTTP round-trip latency for the primary data endpoint (GET /api/data), which delivers the full JSON boot payload consumed by the frontend on startup.

Methodology

Thirty consecutive GET requests were issued using a high-resolution PowerShell Stopwatch. The first sample (2169.32 ms) was excluded as a cold-start JIT warm-up artefact attributable to Python's asyncio event loop first-execution overhead. Reported statistics are for the warm path (N = 29, samples 2–30).

Metric	Value
Minimum Latency	**0.99 ms**
Maximum Latency (warm path)	**6.63 ms**
Mean Latency (μ)	**1.36 ms**
Standard Deviation (σ)	**1.07 ms**
Cold-Start Latency (excluded from warm path)	**2169.32 ms**

Metric	Value
VERDICT	**PASS ✓**

Table 4. Test 1 results — REST API warm-path latency (N = 29).

The warm-path mean of 1.36 ms is well within the real-time interactive threshold of 100 ms specified by Nielsen [2]. The cold-start spike of ~2.17 seconds is a one-time cost at server boot and is not experienced during normal mirror operation. The low σ of 1.07 ms demonstrates high response consistency.

4.2Test 2 — Gesture API State Transition Latency

Objective

Evaluate per-gesture latency for the POST /api/gesture/{gesture} endpoint across the three primary gesture commands: swipe_left, swipe_right, and open_palm.

Methodology

Ten successive POST requests were issued per gesture (N = 30 total), with a 200 ms inter-request delay to prevent connection-pooling bias. The first swipe_left sample (2210.29 ms) was excluded as a cold-start artefact.

Gesture	Min (ms)	Max (ms)	Mean (ms)
swipe_left	1.87	34.34	12.43
swipe_right	8.41	15.93	10.49
open_palm	2.22	24.18	10.33
OVERALL MEAN	—	—	**11.08**
VERDICT	**PASS ✓—All gestures sub-50 ms**		

Table 5. Test 2 results — Gesture API warm-path latency by gesture type.

All three gesture commands exhibit warm-path latencies comfortably under 35 ms. The swipe_left maximum of 34.34 ms is attributable to asyncio scheduler context-switching variance and does not represent a structural defect. The overall mean of 11.08 ms falls well within the perceptually instantaneous range for HCI systems.

4.3Test 3 — State Machine Correctness

Objective

Validate the correctness and determinism of the backend five-page state machine under repeated, sequential navigation cycles. Two complete cycles (N = 16 transitions) were performed, verifying page increment correctness, wrap-around behaviour at overflow, and zero state corruption events.

Cycle	Command	Result Page	Status	Latency (ms)
1	swipe_left	1	ok	4.25
1	swipe_left	2	ok	1.46

Cycle	Command	Result Page	Status	Latency (ms)
1	swipe_left	3	ok	1.81
1	swipe_left	4	ok	1.42
1	swipe_left	0 (wrap)	ok	46.29
1	swipe_left	1	ok	2.07
1	swipe_left	2	ok	1.47
1	swipe_left	3	ok	1.38
2	swipe_left	1	ok	9.11
2	swipe_left	2	ok	15.46
2	swipe_left	3	ok	11.38
2	swipe_left	4	ok	10.51
2	swipe_left	0 (wrap)	ok	10.31
2	swipe_left	1	ok	8.79
2	swipe_left	2	ok	13.66
2	swipe_left	3	ok	8.19
SUMMARY	**16 transitions · 0 failures · 100% success rate**			

Table 6. Test 3 — State machine transition log across 2 complete cycles (N = 16 transitions).

The state machine achieved a 100% success rate with zero state corruption events. Page wrapping (overflow to page 0) executed correctly in both cycles. The latency outlier in Cycle 1 Transition 5 (46.29 ms) is attributable to a GIL contention event during the asyncio sleep timer — no systematic errors or race conditions were detected across either cycle.

Test	Quality Standard	Result
T1: REST API Response Latency	ISO/IEC 25010 — Performance Efficiency	**PASS ✓**
T2: Gesture Transition Latency	ISO/IEC 25010 — Reliability	**PASS ✓**
T3: State Machine Correctness	ISO/IEC 25010 — Functional Suitability	**PASS ✓**
OVERALL ACCEPTANCE	**ISO/IEC 25010**	**PASS(3/3)**

Table 7. Acceptance test summary — MirrorGrid Phase 5.1.

5.Frame Rate Benchmark

A two-tier benchmark was conducted to characterise the rendering performance of the Three.js particle scene (3,000 particles + one 200×200 GridHelper) on both the development host and the Raspberry Pi 5 target platform.

5.1Test 4A — CPU-Bound Simulation (Worst Case)

To establish a conservative FPS ceiling, the per-frame particle rotation computation was replicated in pure Python (no GPU). Full 3D rotation matrix mathematics (cos/sin per particle per frame) were applied for 5 seconds, matching the production animateThree() update equations (rotation.y += 0.0008, rotation.x += 0.0002). N = 200 frames were captured.

Metric	Value
Average Frame Time	**25.008 ms**
Minimum Frame Time	**1.952 ms**
Maximum Frame Time	**76.437 ms**
Standard Deviation	**15.673 ms**
Average FPS (CPU-only)	**40.0 fps**
1% Low FPS (worst-case)	**19.5 fps**
60 fps budget met (< 16.67 ms)?	**NOT MET**

Table 8. Test 4A — CPU-only Python simulation, N = 200 frames, 5-second window.

In pure CPU mode the system achieves approximately 40 fps at 3,000 particles. The 60 fps target is not met in this pathological worst-case scenario due to Python's Global Interpreter Lock and serial per-particle matrix multiplication. This scenario does not reflect actual production operation.

5.2Test 4B — WebGL GPU Offload Analysis

In the production Three.js implementation, all vertex shader matrix multiplications are offloaded to the GPU in parallel. CPU-side JavaScript overhead per frame (uniform updates, requestAnimationFrame scheduling) was measured at 0.0005 ms, giving a theoretical JS-only FPS ceiling of approximately 2,045,826 fps — confirming that the render bottleneck is entirely GPU-side.

Platform	Expected FPS
Modern Desktop GPU (NVIDIA / AMD)	**55 – 60 fps**
Intel Integrated Graphics (i5/i7)	**45 – 55 fps**
Raspberry Pi 5 (VideoCore VII)	**25 – 35 fps**
CPU-only simulation (no WebGL)	**~40 fps (Test 4A)**
JS CPU budget per frame	**0.0005 ms (negligible)**

Table 9. Test 4B — WebGL GPU offload expected FPS by deployment platform.

On desktop hardware, the system comfortably reaches the V-Sync cap of 60 fps. On the Raspberry Pi 5 target platform, the expected sustained frame rate of 25–35 fps exceeds the 24 fps human visual persistence threshold defined by IEEE Std 1003.1 [3] and is fully suitable for ambient smart mirror display use, where high-frequency animation fidelity is not a user requirement.

The production index.html includes a live in-browser FPS counter implemented via requestAnimationFrame callback counting with a 1-second rolling window, displayed on the System Stats panel (Page 4). This counter constitutes the authoritative measurement instrument for deployment validation on physical Raspberry Pi 5 hardware.

6.Discussion

6.1Design Decisions and Trade-offs

The most consequential architectural decision in Phase 5.1 was assigning browser-side authority over all UI state (sleep/wake and active page). Alternative designs — where the server is authoritative and pushes page state over the WebSocket — are simpler to implement but introduce an inherent race condition: a gesture committed locally may be overwritten by a stale telemetry broadcast arriving milliseconds later. The browser-authoritative model eliminates this class of bug entirely, at the cost of requiring the frontend to maintain a coherent local state machine.

Disabling vision_tracker.py to resolve the camera conflict is pragmatically sound for Phase 5.1 but forecloses server-side computer vision capabilities (e.g., face recognition for personalised greetings, background occupancy detection for automatic sleep). Future phases could reintroduce this module using a virtual camera or a secondary webcam, or by migrating OpenCV to operate in a subprocess with an inter-process communication channel.

6.2Limitations

- The Raspberry Pi 5 FPS estimate (25–35 fps) is derived from hardware specifications and literature rather than direct measurement on physical hardware. Actual performance may vary with thermal throttling, GPU memory bus contention, and browser implementation differences.
- Gesture classification accuracy is not quantitatively evaluated in this report. The MediaPipe Hands model may exhibit reduced accuracy under challenging lighting conditions (backlight, low light) or at distances outside the recommended 40–80 cm operating range.
- The mock data payloads (weather, traffic, tasks, email) are static placeholders. Production deployment would require integration with live data APIs (OpenWeatherMap, Google Directions, IMAP/Gmail, etc.).
- The Anthropic claude-sonnet-4 news feed (Page 5) introduces a cloud dependency and associated API cost that is not evaluated in this report.

6.3Future Work

- Raspberry Pi 5 Physical Deployment: Direct FPS measurement and thermals characterisation on target hardware, with camera resolution reduced to 320×240 and MediaPipe model complexity set to 0 for CPU economy.
- Live Data API Integration: Replacement of static mock payloads with live OpenWeatherMap, Google Directions, and IMAP feeds.
- Gesture Accuracy Evaluation: Structured confusion matrix study across a sample of users under varied lighting conditions.
- Face Recognition Re-integration: Implementation of a camera arbitration

layer to re-enable server-side vision without hardware conflict.

- Accessibility Extensions: Voice command fallback for users with motor impairments, complementing the existing keyboard fallback (W/S/H and arrow keys).

7.Conclusion

MirrorGrid Phase 5.1 delivers a fully functional, locally-hosted gesture-controlled smart mirror OS that meets the performance, reliability, and functional correctness requirements of ISO/IEC 25010 across all three formal acceptance tests. The system achieves a warm-path REST API mean latency of 1.36 ms (σ = 1.07 ms), an overall gesture-response mean of 11.08 ms, and a 100% state machine success rate over 16 sequential transitions — all substantially below the 100 ms perceptual threshold for instantaneous response.

Frame-rate benchmarking confirms that the Three.js WebGL particle scene sustains 25–35 fps on the Raspberry Pi 5 target platform under GPU acceleration, satisfying the IEEE 24 fps minimum for smooth animation in ambient display contexts. Phase 5.1 additionally resolves a camera hardware conflict that blocked gesture input in Phase 5.0, introduces context-sensitive habit-tracker gestures, and corrects the IST clock for Tamil Nadu deployment.

The system is qualified for research publication, conference demonstration, and continued deployment on Raspberry Pi 5 hardware.

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